Hamming Codes

A Linear Algebra Tutorial

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A computer represents information as bits. A character, a pixel value, or a sensor reading becomes a string of zeros and ones. The sender transmits those bits, and the receiver wants to recover the same information. However, noise can change a 0 to 1 or a 1 to 0 along the way.

For example, suppose the sender means to send 1101, but the receiver gets 1001. The second bit changed. If every four-bit string is allowed, the receiver cannot tell whether 1001 is the original message or a corrupted one.

To make errors visible, we add check bits and allow only strings that satisfy their constraints. These longer strings are called **codewords**. Detection asks whether the received string violates a constraint. Correction goes further: it tries to recover the transmitted codeword. For correction to work, the possible received strings from different codewords must not overlap under the assumed error bound.

Let k be the number of information bits and n the number of transmitted bits. The $n - k$ extra bits provide the checks:

$$\underbrace{\text{message}}_{k \text{ bits}} \xrightarrow{\text{encode}} \underbrace{\text{codeword}}_{n \text{ bits}} \xrightarrow{\text{possible bit flips}} \underbrace{\text{received word}}_{n \text{ bits}},$$

$$\text{received word} \xrightarrow{\text{decode}} \text{recovered message}.$$

This construction comes from Hamming's 1950 paper.

Prerequisites: vector spaces, linear maps, kernels, bases, invertibility, and rank-nullity. All vector and matrix entries lie in the binary field $\mathbb{F}_2 = \{0, 1\}$. Arithmetic is modulo 2, so $1 + 1 = 0$ and subtraction equals addition.

1 Position Labels and Parity Checks

Choose an integer $m \geq 2$. Let $b_1, \ldots, b_m$ be the standard basis of the label space $\mathbb{F}_2^m$:

$$b_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots, \quad b_m = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Each basis vector has m entries. Every vector in this space has a unique expression

$$a_1 b_1 + a_2 b_2 + \cdots + a_m b_m = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix}, \qquad a_1, \ldots, a_m \in \{0, 1\}.$$

Each coefficient has two choices, so there are 2^m possible labels. Omit the all-zero label and assign the remaining labels to

$$n = 2^m - 1$$

bit positions, using each label once. Call the label at position j the column $h_j \in \mathbb{F}_2^m$. The assignment can use any fixed order. Section 2 explains why the labels must be nonzero and distinct.

A **word** means a whole string of n bits. Number its positions $1, 2, \ldots, n$, and write the bits as a column:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{F}_2^n.$$

This vector has n rows and one column. The entry $x_j \in \{0, 1\}$ is the bit stored at position j. A *coordinate* of the word is simply one such entry. Its label h_j identifies the position and stays fixed when the bit x_j changes.

For the seven-bit example, use the following assignment:

Position j	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Label h_j	001	010	011	100	101	110	111

For example, $h_5 = b_1 + b_3 = \begin{pmatrix} 1 & 0 & 1 \end{pmatrix}^{\mathsf{T}}$. The seven labels are exactly the nonzero combinations of b_1, b_2, b_3.

Later, a compact string such as 101 also denotes the column $\begin{pmatrix} 1 & 0 & 1 \end{pmatrix}^{\mathsf{T}}$. Its leftmost digit is the first entry. This is only a shorter way to write the same vector.

Now place the position labels side by side to form the **parity-check matrix**:

$$H := \underbrace{\begin{pmatrix} | & | & & | \\ h_1 & h_2 & \cdots & h_n \\ | & | & & | \end{pmatrix}}_{m \text{ rows, } n \text{ columns}}. \tag{1.1}$$

Column j belongs to bit position j. Each row specifies one check. Thus multiplying H by a word produces m check results:

$$\underbrace{H}_{m\times n}\,\underbrace{x}_{n\times 1} = \underbrace{Hx}_{m\times 1}.$$

For $m = 3$ and $n = 7$, the multiplication is

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}}_{H:3\times 7} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{pmatrix}}_{x:7\times 1} = \underbrace{\begin{pmatrix} x_4 + x_5 + x_6 + x_7 \\ x_2 + x_3 + x_6 + x_7 \\ x_1 + x_3 + x_5 + x_7 \end{pmatrix}}_{Hx:3\times 1}.$$

A 1 in a row selects a bit; a 0 leaves it out. The selected bits add modulo 2. Their result is 0 for an even number of ones and 1 for an odd number. Each row therefore performs a **parity check**.

We require every check to return zero. The **binary Hamming code** is

$$C := \{c \in \mathbb{F}_2^n : Hc = 0\} = \ker H. \tag{1.2}$$

Here c denotes a valid word, and the zero on the right is an m-entry column of zeros. The encoder sends only words in C.

The columns of H include every basis vector $b_1, \ldots, b_m$. These are m independent columns, and H has only m rows. Hence $\operatorname{rank} H = m$. Rank-nullity now gives

$$\underbrace{n}_{\text{input dimension}} = \underbrace{\operatorname{rank} H}_{m} + \underbrace{\dim \ker H}_{k} \quad \Longrightarrow \quad k = n - m.$$

The m independent equations leave k free binary choices. Equivalently, a basis of C gives k coefficients, each either 0 or 1. Therefore,

$$k = \dim C = n - m, \qquad |C| = 2^k = 2^{n-m}. \tag{1.3}$$

The notation $|C|$ means the number of codewords. For $m = 3$, there are seven transmitted bits, four free information bits, and $2^4 = 16$ codewords. Section 3 shows which four bits we can choose and how the checks determine the remaining three.

2 Syndrome and Single-Error Decoding

Let $c \in C$ be the transmitted word, and let $y \in \mathbb{F}_2^n$ be the received word. Both are $n \times 1$ columns. Define the **error vector** $e \in \mathbb{F}_2^n$ by

$$e = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix}, \qquad e_j := \begin{cases} 0, & y_j = c_j \quad \text{(bit unchanged)}, \\ 1, & y_j \neq c_j \quad \text{(bit flipped)}. \end{cases}$$

Adding 0 preserves a bit. Adding 1 flips it, because $0 + 1 = 1$ and $1 + 1 = 0$. Consequently,

$$y = \begin{pmatrix} c_1 + e_1 \\ c_2 + e_2 \\ \vdots \\ c_n + e_n \end{pmatrix} = c + e.$$

The receiver can calculate Hy without knowing c. Call this $m \times 1$ vector the **syndrome**, denoted by s:

$$\boxed{s := Hy = H(c + e) = \underbrace{Hc}_{0} + He = He.} \tag{2.1}$$

Each entry records one parity-check result. The valid transmitted word contributes zero, so only the error affects the syndrome.

To describe a single-bit error, let $u_j \in \mathbb{F}_2^n$ have a 1 in row j and zeros elsewhere:

$$u_j = \underbrace{\begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}}_{n \times 1}.$$

Multiplying by this vector gives coefficient 1 to column j and coefficient 0 to every other column:

$$Hu_j = 0h_1 + \cdots + 1h_j + \cdots + 0h_n = h_j.$$

Assume at most one bit flips. There are only two cases:

$$\begin{aligned} e = 0 \quad &\implies \quad s = 0, \\ e = u_j \quad &\implies \quad s = h_j \neq 0. \end{aligned}$$

Every label is nonzero. Therefore, under this error bound,

$$s = 0 \iff e = 0.$$

The labels are also distinct, so $s = h_j$ identifies exactly one position. Let $\widehat{c}$ denote the decoder's estimate of the transmitted word. The decoding rule is

$$\widehat{c} = \begin{cases} y, & s = 0, \\ y + u_j, & s = h_j. \end{cases} \tag{2.2}$$

In the second case, adding u_j flips the same bit back:

$$y + u_j = c + u_j + u_j = c.$$

This also explains our choice of labels. A zero label would hide a single-bit error, and repeated labels would confuse two positions:

$$\begin{aligned} h_j = 0 \quad &\implies \quad Hu_j = H0, \\ h_i = h_j \ (i \neq j) \quad &\implies \quad Hu_i = Hu_j. \end{aligned}$$

We therefore need different syndromes for $0, u_1, \ldots, u_n$: one for no error and one for each possible flipped position. An m-bit syndrome has 2^m possible values, so

$$n + 1 \leq 2^m.$$

The Hamming construction uses all of them. This gives the largest possible length $n = 2^m - 1$ for m checks that distinguish these single-error cases.

3 Encoding and Choosing Check Positions

Choose a set P of m check positions. Let D be the set of the remaining $k = n - m$ data positions. Within each set, list the positions in increasing order. Extract the corresponding columns of H:

$$\underbrace{H_P}_{m\times m} : \text{columns at check positions}, \qquad \underbrace{H_D}_{m\times k} : \text{columns at data positions}.$$

Let $p \in \mathbb{F}_2^m$ contain the unknown check bits, and let $d \in \mathbb{F}_2^k$ contain the chosen information bits. Both are column vectors. Group the check positions first for this calculation:

$$\underbrace{\begin{pmatrix} H_P & H_D \end{pmatrix}}_{m\times n} \underbrace{\begin{pmatrix} p \\ d \end{pmatrix}}_{n\times 1} = \underbrace{H_P p}_{m\times 1} + \underbrace{H_D d}_{m\times 1} = 0.$$

The two matrix-vector products must add to zero. Move $H_D d$ to the right:

$$H_P p + H_D d = 0 \quad \Longrightarrow \quad H_P p = -H_D d = H_D d.$$

The last equality uses $-0 = 0$ and $-1 = 1$ in $\mathbb{F}_2$. In particular, $1 + 1 = 0$ makes each entry its own additive inverse.

For example, with $P = \{1, 2, 4\}$ and $D = \{3, 5, 6, 7\}$, the regrouping is

$$\underbrace{\begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \end{pmatrix}}_{\text{transmitted order}} \longleftrightarrow \underbrace{\left(\begin{array}{c} c_1 \\ c_2 \\ c_4 \\ \hline c_3 \\ c_5 \\ c_6 \\ c_7 \end{array}\right)}_{\text{grouped order}} = \begin{pmatrix} p \\ d \end{pmatrix}.$$

Reorder the columns of H in the same way:

$$[\, h_1 \; h_2 \; h_4 \mid h_3 \; h_5 \; h_6 \; h_7 \,] = [\, H_P \; H_D \,].$$

Each bit is still multiplied by its own column. After solving for p, put the bits back in transmitted order.

For each chosen data vector, the equation to solve is

$$\underbrace{H_P}_{\text{known}} \qquad \underbrace{p}_{\text{unknown}} = \underbrace{H_D d}_{\text{known}}.$$

Encoding requires one solution for every choice of data:

$$\boxed{\begin{array}{c} \text{For every } d \in \mathbb{F}_2^k,\ H_P p = H_D d \\ \text{has exactly one solution } p \in \mathbb{F}_2^m \\ \Longleftrightarrow \quad H_P \text{ is invertible.} \end{array}} \tag{3.1}$$

($\Leftarrow$) If H_P is invertible, multiply both sides by its inverse:

$$p = \underbrace{H_P^{-1}}_{m\times m}\underbrace{H_D}_{m\times k}\underbrace{d}_{k\times 1}.$$

This gives one solution for every d, and invertibility makes it unique.

($\Rightarrow$) Set $d = 0$. Then $p = 0$ solves the equation. Uniqueness rules out any other solution, so

$$\ker H_P = \{0\} \Longrightarrow \operatorname{rank} H_P = m \Longrightarrow H_P \text{ is invertible.}$$

Here H_P is square. Its m columns must therefore form a basis of $\mathbb{F}_2^m$.

Why an arbitrary choice can fail. For $m = 3$, choose $P = \{3, 5, 6\}$. The selected labels give

$$\underbrace{\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}}_{H_P} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

For the all-zero data vector, both the all-zero and all-one check vectors work. Thus the information bits do not determine a unique codeword with this choice of positions.

One complete encoding and decoding example. Use $P = \{1, 2, 4\}$ and $D = \{3, 5, 6, 7\}$ from the regrouping above. The equation $H_P p = H_D d$ becomes

$$\underbrace{\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{H_P:3\times 3}\underbrace{\begin{pmatrix} c_1 \\ c_2 \\ c_4 \end{pmatrix}}_{p:3\times 1} = \underbrace{\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}}_{H_D:3\times 4}\underbrace{\begin{pmatrix} c_3 \\ c_5 \\ c_6 \\ c_7 \end{pmatrix}}_{d:4\times 1}.$$

Reading the three rows gives

$$\begin{aligned} c_4 &= c_5 + c_6 + c_7, \\ c_2 &= c_3 + c_6 + c_7, \\ c_1 &= c_3 + c_5 + c_7. \end{aligned}$$

For the message below, the check bits are

$$d = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \qquad p = \begin{pmatrix} 1+0+1 \\ 1+1+1 \\ 0+1+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Put these entries into their designated positions. Then flip bit 6 during transmission:

$$c = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \qquad y = c + u_6 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

The receiver evaluates the three checks:

$$Hy = \begin{pmatrix} 0+0+0+1 \\ 1+1+0+1 \\ 0+1+0+1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = h_6.$$

So the decoder flips bit 6 back, giving $\widehat{c} = y + u_6 = c$. Reading the data positions recovers

$$\begin{pmatrix} \widehat{c}_3 \\ \widehat{c}_5 \\ \widehat{c}_6 \\ \widehat{c}_7 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} = d.$$

In general, call the encoding map E. It places $H_P^{-1} H_D d$ in the check positions and d in the data positions:

$$E : \mathbb{F}_2^k \longrightarrow C.$$

This map is linear. Its inverse reads the data positions, since the checks uniquely determine the other entries. Thus E is an isomorphism. A different valid choice of P changes how we encode the message; it leaves $C = \ker H$ unchanged.

4 Ambiguity, Weight, and Minimum Distance

The single-error rule works because one flipped bit selects one column of H. What changes when two bits flip?

Choose distinct positions i, j. The error is $e = u_i + u_j$, so

$$s = H(u_i + u_j) = \underbrace{[\, h_i \; h_j \,]}_{m \times 2} \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{2 \times 1} = h_i + h_j.$$

The two columns are distinct, so their sum is nonzero. The Hamming matrix contains every nonzero m-bit column. Thus there is a position r with

$$h_i + h_j = h_r, \qquad H(u_i + u_j) = H u_r.$$

Here $r \neq i, j$: either equality would force one of h_i, h_j to be zero.

The same syndrome now has two explanations: bits i, j flipped, or bit r flipped. The single-error decoder chooses the second explanation. If bits i, j actually flipped, its output is

$$\widehat{c} = c + u_i + u_j + u_r \neq c, \qquad H\widehat{c} = h_i + h_j + h_r = 0.$$

It returns a different valid codeword. The map $e \mapsto He$ is therefore not injective when both one-bit and two-bit errors are allowed: the two different error vectors above have the same image.

The encoding map $E : \mathbb{F}_2^k \to C$ remains an isomorphism. It assigns each message a different codeword. The ambiguity concerns which error changed the transmitted word. Since $H\widehat{c} = 0$, the incorrect result still passes the parity checks.

Counting changed positions

For a binary column vector

$$v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{F}_2^n,$$

its **support** lists the positions containing a 1. Its **weight** counts those positions:

$$\operatorname{supp}(v) := \{j \in \{1, \dots, n\} : v_j = 1\}, \qquad \operatorname{wt}(v) := |\operatorname{supp}(v)|.$$

Here $|A|$ means the number of elements in a finite set A.

Let $x, z \in \mathbb{F}_2^n$ be two words. Their difference records exactly where they disagree:

$$z - x = \begin{pmatrix} z_1 - x_1 \\ \vdots \\ z_n - x_n \end{pmatrix}, \qquad z_j - x_j = \begin{cases} 0, & z_j = x_j, \\ 1, & z_j \neq x_j. \end{cases}$$

The **Hamming distance** is the number of differing positions:

$$d_H(x, z) := |\{j : x_j \neq z_j\}| = \operatorname{wt}(z - x).$$

In particular, $y - c = e$ gives

$$d_H(c, y) = \operatorname{wt}(e).$$

Thus $\operatorname{wt}(e)$ counts flipped bits. By contrast, $\operatorname{wt}(He)$ counts failed checks. One flip can fail several checks, so these counts need not agree.

We will also use the triangle inequality. Choose any intermediate word $w \in \mathbb{F}_2^n$. At a position where x and z differ, at least one of the pairs (x, w) and (w, z) must differ. Counting these positions gives

$$d_H(x, z) \leq d_H(x, w) + d_H(w, z).$$

Undetectable errors and the closest codewords

Define the set of **nonzero undetectable errors** by

$$\mathcal{U} := \{e \in \mathbb{F}_2^n : e \neq 0,\ He = 0\} = \ker H \setminus \{0\} = C \setminus \{0\}.$$

If $c \in C$ and $e \in \mathcal{U}$, then

$$y = c + e \neq c, \qquad Hy = Hc + He = 0 + 0 = 0.$$

The word changes, but all checks still pass. A nonzero codeword can therefore also serve as an undetectable error; these names describe different roles of the same vector.

The **minimum distance** $d_{\min}$ is the smallest distance between two distinct codewords. Take $c, c' \in C$ with $c \neq c'$. Linearity lets us move the first codeword to zero:

$$(c, c') \ \longmapsto \ (c - c, c' - c) = (0, v), \qquad v := c' - c.$$

Subtracting the same word preserves the differing positions. Also, v is a nonzero codeword because C is a subspace. Hence

$$d_H(c, c') = d_H(0, v) = \mathrm{wt}(v).$$

Conversely, every nonzero codeword v occurs in the pair $(0, v)$. Thus the two minimizations examine the same distances:

$$\boxed{\begin{aligned} d_{\min} &:= \min_{\substack{c,c' \in C \\ c' \neq c}} d_H(c, c') \\ &= \min_{\substack{v \in C \\ v \neq 0}} \mathrm{wt}(v) = \min_{\substack{e \neq 0 \\ He = 0}} \mathrm{wt}(e). \end{aligned}} \tag{4.1}$$

This identity holds for any binary linear code with at least two codewords. It identifies the smallest undetectable error. Therefore,

$$1 \leq \mathrm{wt}(e) < d_{\min} \quad \Longrightarrow \quad He \neq 0.$$

Every nonzero error of weight at most $d_{\min} - 1$ is detected. At weight $d_{\min}$, an undetectable error exists: choose a minimum-weight nonzero codeword as e. Other errors of that weight may still be detected.

5 Minimum Distance from the Columns of H

Equation (4.1) reduces the problem to finding the fewest flipped positions that make $He = 0$. Matrix multiplication shows what those positions do:

$$He = \underbrace{[\, h_1 \ \cdots \ h_n \,]}_{m \times n} \underbrace{\begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}}_{n \times 1} = e_1 h_1 + \cdots + e_n h_n.$$

Each $e_j = 0$ removes column j; each $e_j = 1$ includes it. The result is an $m \times 1$ column.

For a nonzero error e, list its flipped positions as

$$I = \mathrm{supp}(e) = \{j_1, \ldots, j_r\}, \qquad j_1 < \cdots < j_r, \qquad r = \mathrm{wt}(e).$$

Let H_I be the $m \times r$ matrix containing those columns, in that order. Let $\mathbf{1}_r$ be the $r \times 1$ column of ones. Then

$$He = H_I \mathbf{1}_r = \underbrace{[\, h_{j_1} \ \cdots \ h_{j_r} \,]}_{m \times r} \underbrace{\begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}}_{r \times 1}.$$

Thus $He = 0$ means that the selected columns cancel. Since $\mathbf{1}_r \neq 0$, this is a nontrivial linear relation.

Let ℓ be the smallest number of columns of H that form a linearly dependent set. We will show

$$\boxed{d_{\min} = \ell.} \tag{5.1}$$

The argument applies to any binary check matrix whose kernel contains a nonzero vector.

($\ell \le d_{\min}$)**.** Choose a nonzero codeword v of minimum weight. Its support I has $d_{\min}$ positions. The corresponding columns satisfy

$$\underbrace{H_I}_{m\times d_{\min}} \underbrace{\mathbf{1}_{d_{\min}}}_{d_{\min}\times 1} = Hv = 0.$$

These $d_{\min}$ columns are dependent. The smallest dependent set cannot be larger, so $\ell \le d_{\min}$.

($d_{\min} \le \ell$)**.** Choose ℓ dependent columns, at distinct positions $j_1, \ldots, j_\ell$. By linear dependence, there is a nonzero coefficient column $a \in \mathbb{F}_2^\ell$ with

$$\underbrace{[\, h_{j_1} \ \cdots \ h_{j_\ell} \,]}_{m\times \ell} \underbrace{\begin{pmatrix} a_1 \\ \vdots \\ a_\ell \end{pmatrix}}_{\ell\times 1} = 0, \qquad a \neq 0.$$

Build an $n \times 1$ error column e by putting a_r at position j_r, for $r = 1, \ldots, \ell$. Put 0 in every other position. This places the same coefficients beside the same columns in He, so

$$He = [\, h_{j_1} \ \cdots \ h_{j_\ell} \,]a = 0, \qquad e \neq 0.$$

At most ℓ entries of e are nonzero. Hence

$$d_{\min} \le \mathrm{wt}(e) \le \ell.$$

Applying the column test to Hamming codes

One column is dependent only when it is zero. For two distinct nonzero binary columns, the only remaining nonzero coefficient choice that could give zero is

$$[\, h_i \ h_j \,] \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0 \quad \Longleftrightarrow \quad h_i = h_j.$$

The Hamming matrix has no zero or repeated columns. Therefore neither one nor two columns can be dependent, and $d_{\min} \ge 3$.

Three dependent columns exist because H contains all nonzero labels. Choose the labels b_m, b_{m-1}, and $b_m + b_{m-1}$, and call their distinct positions i, j, r. On the final two rows, the corresponding columns give

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0+1+1 \\ 1+0+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

All earlier rows are zero. Thus

$$[\, h_i \ h_j \ h_r \,] \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 0 \quad \Longrightarrow \quad d_{\min} \le 3.$$

Combining the two bounds gives

$$\boxed{d_{\min} = 3.} \tag{5.2}$$

Every one-bit or two-bit error is detected. The error $u_i + u_j + u_r$ from these three columns is undetectable. Detecting a two-bit error does not identify its two positions: the syndrome collision in Section 4 still applies. We next determine when every allowed error has a unique explanation.

6 Correction Radius and Hamming Balls

The single-error decoder works because different allowed errors have different syndromes. We now ask when the same idea works for more errors.

Let $t \in \{0, \dots, n\}$ be the maximum number of allowed bit flips. Collect the allowed error vectors in

$$\mathcal{E}_t := \{e \in \mathbb{F}_2^n : \mathrm{wt}(e) \le t\}.$$

This includes $e = 0$, since a transmission may have no error. Given a received word y, find an error vector satisfying

$$e \in \mathcal{E}_t, \qquad He = Hy.$$

Under the error bound, the actual error satisfies both conditions. Each solution gives a candidate codeword $c = y - e = y + e$. If the allowed solution is unique, it must be the actual error.

For unique correction, there must be only one allowed solution. In other words, different vectors in $\mathcal{E}_t$ must have different syndromes. Here is why a collision prevents correction. Suppose

$$e, e' \in \mathcal{E}_t, \qquad e \ne e', \qquad He = He' = Hy.$$

Then the two candidate codewords satisfy

$$\begin{aligned} c &:= y - e, \quad Hc = Hy - He = 0, \\ c' &:= y - e', \quad Hc' = Hy - He' = 0. \end{aligned}$$

They are distinct because $e \ne e'$. Yet both explain the received word:

$$y = c + e = c' + e'.$$

Both explanations obey the error bound, so the receiver cannot choose between them with certainty.

The correction threshold

Two allowed errors can differ in at most $2t$ positions. This is why the distance between codewords must exceed $2t$:

$$\boxed{\begin{array}{c}\text{Every error of weight at most } t \text{ is uniquely correctable} \\ \iff \quad d_{\min} > 2t.\end{array}} \tag{6.1}$$

($\Leftarrow$) Assume $d_{\min} > 2t$. Suppose two distinct allowed errors have the same syndrome. Then

$$H(e + e') = He + He' = 0, \qquad e + e' \ne 0.$$

Thus $e + e'$ is a nonzero codeword. A nonzero entry of $e + e'$ must occur at a position used by e or e'. Therefore,

$$d_{\min} \le \mathrm{wt}(e + e') \le \mathrm{wt}(e) + \mathrm{wt}(e') \le 2t.$$

This contradicts $d_{\min} > 2t$. All allowed errors have distinct syndromes, so correction is unique.

($\Rightarrow$) We prove the contrapositive. Suppose $d_{\min} \le 2t$. Choose a nonzero codeword v of weight $w = d_{\min}$. We will split its w ones between two allowed error vectors.

Put $r = \min(w, t)$. Assign r of the ones to e and the remaining $w - r$ ones to e'. Both groups have at most t entries because $w \le 2t$.

For the next display only, list these two groups of positions first and the zero positions last. Use the same position order in all three vectors. For a block length $q > 0$, write

$$\mathbf{1}_q := \underbrace{\begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}}_{q\times 1}, \qquad \mathbf{0}_q := \underbrace{\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}}_{q\times 1}.$$

A block of length zero is omitted. With these blocks, the split becomes

$$\underbrace{\begin{pmatrix} \mathbf{1}_r \\ \mathbf{1}_{w-r} \\ \mathbf{0}_{n-w} \end{pmatrix}}_{v} = \underbrace{\begin{pmatrix} \mathbf{1}_r \\ \mathbf{0}_{w-r} \\ \mathbf{0}_{n-w} \end{pmatrix}}_{e} + \underbrace{\begin{pmatrix} \mathbf{0}_r \\ \mathbf{1}_{w-r} \\ \mathbf{0}_{n-w} \end{pmatrix}}_{e'}.$$

Each column has $r + (w - r) + (n - w) = n$ entries. In particular,

$$v = e + e', \qquad \mathrm{wt}(e) = r \le t, \qquad \mathrm{wt}(e') = w - r \le t.$$

Now transmit either 0 or v. Since adding an error twice cancels it,

$$\underbrace{0 + e}_{\text{sent } 0} = e = (e + e') + e' = \underbrace{v + e'}_{\text{sent } v}.$$

The same word arrives from two distinct codewords, using at most t flips in either case. Unique correction is impossible.

Since $d_{\min}$ and t are integers, $2t < d_{\min}$ means $2t \le d_{\min} - 1$. The largest guaranteed number of correctable flips is therefore

$$\boxed{t_{\max} = \left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor}. \tag{6.2}$$

Here $\lfloor x \rfloor$ means rounding down to the nearest integer. For a Hamming code, $d_{\min} = 3$ gives $t_{\max} = 1$.

Balls collect the possible received words

Fix a transmitted codeword c. The **closed Hamming ball** with center c and radius t is

$$B_t(c) := \{y \in \mathbb{F}_2^n : d_H(c, y) \le t\} = \{c + e : e \in \mathcal{E}_t\}. \tag{6.3}$$

The center is the transmitted word. The radius is the maximum number of flips. Thus $B_t(c)$ contains every word that could arrive from c under this error bound. A member of the ball need not itself be a codeword.

If balls around two different codewords overlap, a word in the overlap has two possible origins. In symbols, $y \in B_t(c) \cap B_t(c')$ means that there are error vectors e, e' satisfying

$$\begin{cases} y = c + e = c' + e', \\ \mathrm{wt}(e) \le t, \quad \mathrm{wt}(e') \le t. \end{cases}$$

So unique correction means that the balls around distinct codewords are disjoint. By (6.1), this happens exactly when $d_{\min} > 2t$.

Why the assumed error bound matters

Take the three-bit repetition code $C = \{000, 111\}$. It encodes one bit by repeating it three times. This is also the $m = 2$ Hamming code. With at most one flip, the possible received words are

$$B_1(000) = \{000, 100, 010, 001\},$$
$$B_1(111) = \{111, 011, 101, 110\}.$$

The balls are disjoint. Receiving 100 identifies 000, because

$$d_H(000, 100) = 1, \qquad d_H(111, 100) = 2 > 1.$$

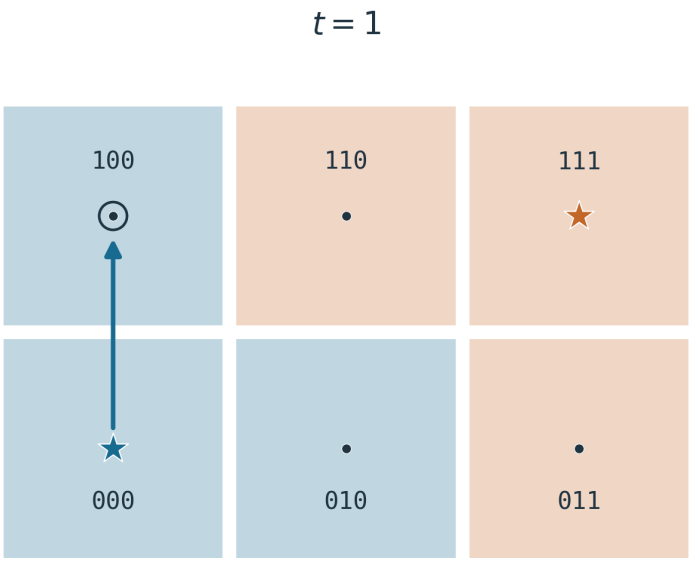


Figure 1: Stars mark the two codewords. Tile colors show ball membership. The ringed word 100 belongs only to the blue ball. Each horizontal or vertical grid step flips one bit. This grid shows six of the eight three-bit words; 001 and 101 are omitted. Each full ball contains four words. Appendix A explains the grid positions.

Now allow two flips. The same received word has two permitted explanations:

$$100 = 000 + \underbrace{100}_{\text{one flip}} = 111 + \underbrace{011}_{\text{two flips}}.$$

Thus $100 \in B_2(000) \cap B_2(111)$. Choosing the nearer codeword gives 000, but the transmitted codeword could have been 111.

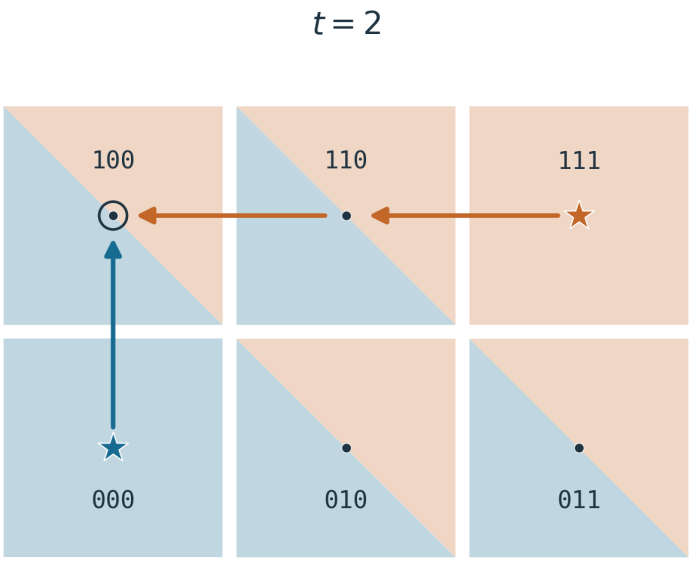


Figure 2: The same grid with radius two. Split tiles belong to both balls. The arrows show two paths to 100. Each full ball contains seven words; 001 and 101 remain omitted from the grid.

7 Counting Errors and the Hamming Bound

To compare the sizes of these balls, first count the words at one fixed distance. For $i \in \{0, \dots, n\}$, define the **Hamming sphere**

$$S_i(c) := \{y \in \mathbb{F}_2^n : d_H(c, y) = i\}.$$

A sphere uses exactly i flips. A ball includes every layer from zero flips up to its radius.

Choose a set $I \subseteq \{1, \dots, n\}$ of exactly i positions to flip. Define the error entries by

$$e_j = \begin{cases} 1, & j \in I, \\ 0, & j \notin I. \end{cases}$$

These entries form an n-entry error column. Adding it to the transmitted column gives

$$y = c + e = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} c_1 + e_1 \\ c_2 + e_2 \\ \vdots \\ c_n + e_n \end{pmatrix}.$$

At a selected position, $c_j + 1$ is the opposite bit. At every other position, $c_j + 0 = c_j$. Thus each choice of I produces exactly one word at distance i.

Conversely, the changed positions can be read from that word:

$$I = \{j : y_j \neq c_j\}.$$

Different choices of positions therefore give different received words. The number of choices is the binomial coefficient $\binom{n}{i}$: the number of ways to choose i positions from n, without regard to order. Hence

$$|S_i(c)| = \binom{n}{i}.$$

There is one choice for zero flips, n choices for one flip, $\binom{n}{2}$ choices for two flips, and so on. A word cannot belong to two different distance layers. Adding their sizes gives

$$\boxed{V(n,t) := |B_t(c)| = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{t}.} \tag{7.1}$$

This count starts with $1 + n + \binom{n}{2} + \cdots$ and stops at the layer for t flips. The notation $V(n,t)$ is called the ball's **volume**, but it simply counts words. It does not depend on the center c. In particular,

$$V(n,0) = 1, \qquad V(n,1) = 1 + n, \qquad V(n,n) = 2^n.$$

The last equality holds because allowing all n positions to flip can produce any binary word of length n.

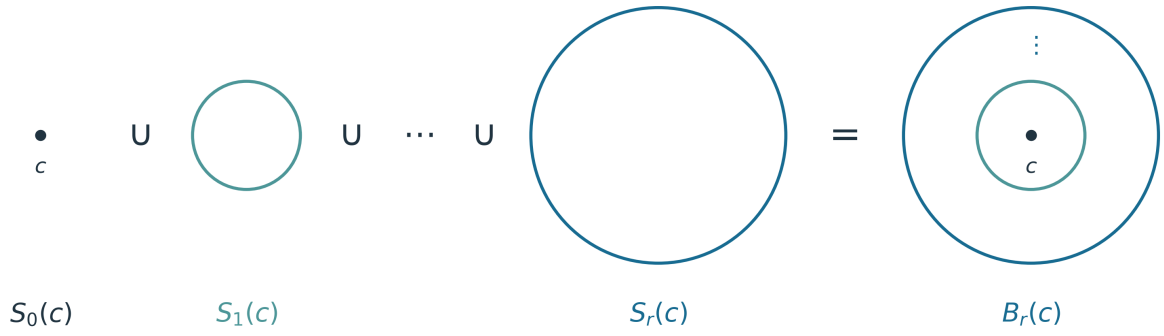


Figure 3: A ball of radius r consists of its center and the layers $S_1(c), \ldots, S_r(c)$. The circles represent discrete distance layers. Dots indicate omitted intermediate layers.

If a code corrects every error of weight at most t, its radius-t balls are disjoint. There are $|C|$ codeword centers, and each ball contains $V(n,t)$ words. The whole space contains 2^n words, since each of the n bits has two choices. Therefore,

$$\boxed{\underbrace{|C|}_{\text{number of balls}} \quad \underbrace{V(n,t)}_{\text{words per ball}} \quad \leq \quad \underbrace{2^n}_{\text{all received words}}.} \tag{7.2}$$

This is the **Hamming bound**.

For a linear code with m independent checks, H has rank m. Its kernel has dimension $n - m$, so $|C| = 2^{n-m}$. Substituting into (7.2) gives

$$2^{n-m} V(n,t) \leq 2^n \quad \Longrightarrow \quad V(n,t) \leq 2^m.$$

The same bound follows by counting syndromes: an m-entry binary syndrome has 2^m possible values, and these must distinguish all $V(n,t)$ allowed errors. Taking base-two logarithms yields

$$m \geq \lceil \log_2 V(n,t) \rceil .$$

The ceiling $\lceil x \rceil$ rounds up to the nearest integer, because the number of checks must be an integer. Having enough syndromes is necessary; the chosen matrix must also assign different syndromes to different allowed errors.

A code is **perfect** for radius t when its disjoint radius-t balls cover the whole space. For Hamming codes, $d_{\min} = 3$ makes the radius-one balls disjoint. Also, $n = 2^m - 1$, so

$$|C|V(n,1) = 2^{n-m}(1+n) = 2^{n-m}2^m = 2^n. \tag{7.3}$$

The disjoint balls already contain as many words as the whole space. They must cover it, so Hamming codes are perfect for radius one.

8 Information Bits and Correction Radius

In the Hamming family, increasing m increases both the word length and the number of information bits:

Check bits m	**Word length** $n = 2^m - 1$	**Information bits** $k = n - m$
2	3	1
3	7	4
4	15	11
5	31	26

However, the minimum distance remains 3, so the guaranteed correction radius remains one. Longer words in this family carry more information; they do not guarantee correction of more flips.

To see a different tradeoff, hold the word length fixed at seven. A seven-bit **repetition code** encodes one information bit $b \in \mathbb{F}_2$ by sending

$$b \quad \longmapsto \quad c = \begin{pmatrix} b \\ b \\ b \\ b \\ b \\ b \\ b \end{pmatrix}.$$

The two possible inputs give

$$C_{\text{rep}} = \{0000000, 1111111\}.$$

Only the first bit is free. Each of the other six must equal it, giving six independent checks:

$$c_1 + c_2 = 0, \quad c_1 + c_3 = 0, \quad \ldots, \quad c_1 + c_7 = 0.$$

Thus $k = 1$ and $m = 6$. The two codewords differ in all seven positions, so $d_{\min} = 7$. Applying the detection and correction formulas gives

Seven-bit code	**Information bits** k	**Check bits** m	$d_{\min}$	**Guaranteed detection**	**Guaranteed correction**
Hamming	4	3	3	up to 2 flips	up to 1 flip
Repetition	1	6	7	up to 6 flips	up to 3 flips

The repetition code carries fewer information bits, but its codewords are farther apart. Each codeword can therefore have a larger correction ball. At the respective guaranteed radii,

$$\begin{aligned} \text{Hamming:} \quad & 16\,V(7,1) = 16(1+7) = 128, \\ \text{Repetition:} \quad & 2\,V(7,3) = 2(1+7+21+35) = 128. \end{aligned}$$

Both families of balls cover all $2^7 = 128$ words without overlap. Hamming uses sixteen smaller correction regions; repetition uses two larger ones.

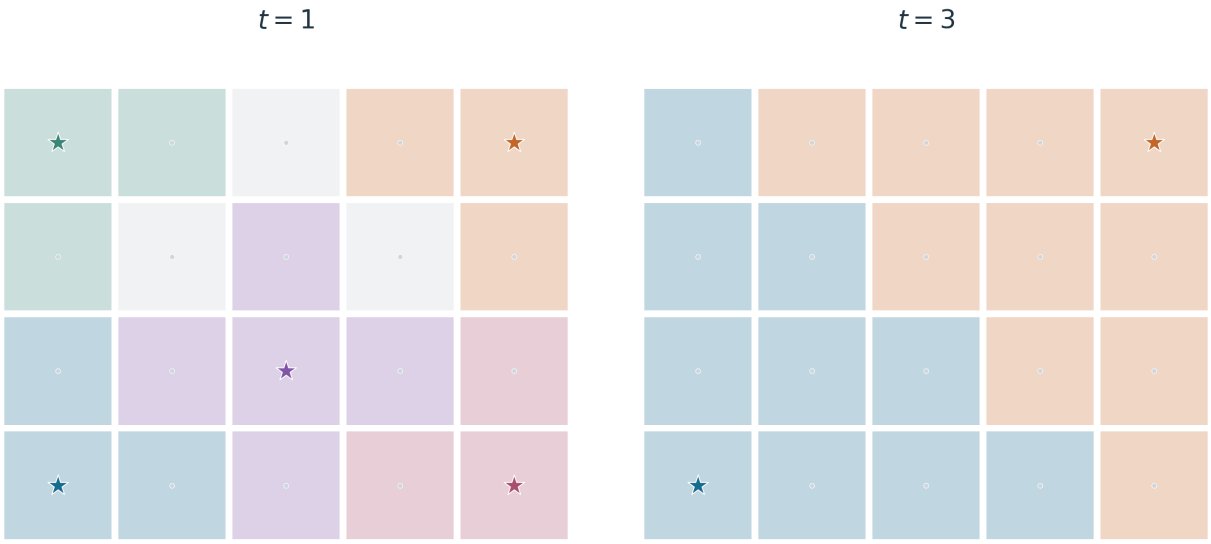


Figure 4: The same 20-word subset of $\mathbb{F}_2^7$ is shown in both panels. The left panel shows Hamming balls of radius one; the right shows repetition-code balls of radius three. Stars mark codewords, and colors identify correction regions. Grey words have Hamming-code centers outside the displayed subset. Appendix A defines the grid.

A Reading the Grid Figures

The grids show selected binary words in a two-dimensional arrangement. To define the arrangement, split the n bit positions into two groups. The first group has $a = \lfloor n/2 \rfloor$ positions; the second has $n - a$ positions.

Let i be the horizontal grid coordinate and j the vertical coordinate, starting from $(0, 0)$ at the bottom-left corner. In the first group, turn on the first j bits. In the second group, turn on the first i bits. This gives the column vector

$$v_{i,j} := \begin{pmatrix} \mathbf{1}_j \\ \mathbf{0}_{a-j} \\ \mathbf{1}_i \\ \mathbf{0}_{n-a-i} \end{pmatrix}, \qquad 0 \le i \le n - a, \quad 0 \le j \le a.$$

Here $\mathbf{1}_q$ is a column of q ones and $\mathbf{0}_q$ is a column of q zeros. Empty blocks are omitted. The four block lengths add to

$$j + (a - j) + i + (n - a - i) = n.$$

Place this word at column i, row j. Moving right increases i by one, so it flips the next zero in the second group to one. Moving up increases j by one, so it flips the next zero in the first group to one. Each grid step therefore changes exactly one bit.

Between $v_{i,j}$ and $v_{i',j'}$, the first groups differ in $|j - j'|$ positions and the second groups differ in $|i - i'|$ positions. Since the two groups use separate positions,

$$d_H(v_{i,j}, v_{i',j'}) = |i - i'| + |j - j'|.$$

The right-hand side is the number of horizontal and vertical steps in a shortest grid path, also called Manhattan distance. Thus grid distance equals Hamming distance for the displayed words.

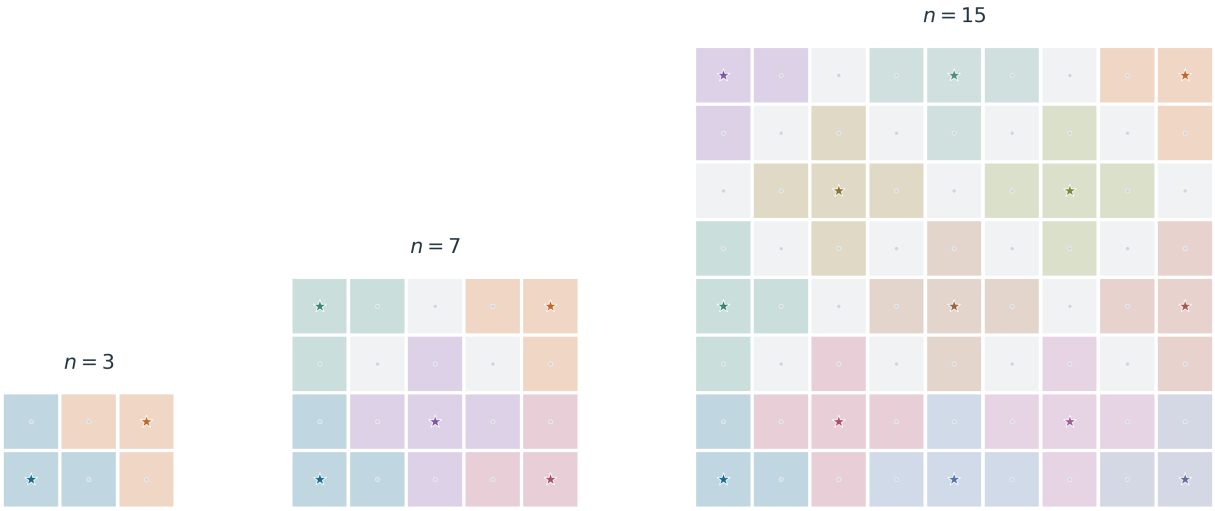


Figure A1: Radius-one correction regions at three word lengths. All panels use the same tile size. Stars mark codewords, and colors identify their balls. Grey words belong to balls with centers outside the grid.

The grid has $(n - a + 1)(a + 1)$ positions, so it shows only part of the full space:

Word length n	**Displayed words**	**Full space size** 2^n
3	6	8
7	20	128
15	72	32,768

The full graph uses all words of $\mathbb{F}_2^n$ as vertices and connects two words whenever they differ in one bit. This is the hypercube graph Q_n. A shortest path flips each differing position once, so its length is d_H. Its graph balls are therefore Hamming balls.